SpotOn: An AI-Powered Multi-Camera Person Tracking System for Real-Time Situational Awareness

1st Krittin Setdhavanich
Department of Computer Engineering
Kasetsart University
Bangkok, Thailand
krittin.set@ku.th

2nd Yanatchara Jeraja
Department of Computer Engineering
Kasetsart University
Bangkok, Thailand
yanatchara.j@ku.th

3rd Supaporn Erjongmanee
Department of Computer Engineering
Kasetsart University
Bangkok, Thailand
supaporn.e@ku.th

4th Chaiwat Klampol
Department of Computer Engineering
Kasetsart University
Bangkok, Thailand
fengcwp@ku.ac.th

***Abstract*—Tracking individuals across multiple cameras is essential for surveillance but remains challenging due to lighting changes, varying viewpoints, and occlusions. In this paper, we propose SpotOn, a real-time multi-camera person tracking system that maintains identity continuity across distributed camera networks. The system uses a CNN-based object detector for person detection, a tracking-by-detection algorithm for intra-camera tracking, and an appearance-based re-identification (Re-ID) network for cross-camera identity matching. A homography-based spatial matching module projects each detection onto a shared 2D floor plan, allowing operators to monitor all tracked individuals from a bird's-eye view. For cameras with overlapping fields of view, identities are resolved geometrically; Re-ID is invoked only for non-overlapping blind spots, reducing computational cost. We evaluated the system on the MTMMC dataset [8] across a university campus and an industrial factory, achieving an average IDF1 of 61.8% and MOTA of 54.3% with 49.4 ms per-frame latency.**

***Index Terms*—Multi-Camera Tracking, Person Re-Identification, Computer Vision, Spatial Matching, Surveillance Systems**

I. Introduction

In large-scale environments such as university campuses and factories, tracking individuals across multiple cameras is essential for security and operational efficiency. When a person moves from one camera view to another, the system must re-establish their identity despite changes in lighting, perspective, and possible occlusion. Without an automated solution, security operators must manually match individuals across fragmented video feeds, which is time-consuming and error-prone.

Existing multi-camera tracking systems often rely entirely on appearance-based re-identification (Re-ID) for cross-camera association. This approach is computationally expensive and performs poorly in scenarios where subjects wear similar clothing, as is common in factory environments.

In this paper, we propose SpotOn, a real-time multi-camera person tracking system that uses a two-tier matching strategy. For cameras with overlapping fields of view, identities are resolved geometrically using homography-based spatial matching, which is faster and more robust than appearance-based Re-ID in such zones. For non-overlapping blind spots, the Re-ID module is activated only when a person approaches the edge margin of a camera frame. This design reduces overall computational cost while maintaining tracking accuracy.

The main contributions of this work are: (1) a two-tier identity matching strategy that prioritizes geometry-based matching for overlapping views and invokes Re-ID only at camera edge boundaries; (2) a homography-based spatial matching module that projects detections onto a shared 2D floor plan and resolves identities without appearance features; and (3) an evaluation on the MTMMC dataset [8] demonstrating competitive IDF1, MOTA, and sub-second per-frame latency.

II. Related Works

Multi-camera person tracking requires solving several sub-problems: detecting persons in each frame, tracking them within a single camera, and re-establishing identities across cameras.

A. Foundational Components

For object detection, we use YOLOv26m [9], a CNN-based detector with an NMS-free decoupled head that eliminates the post-processing bottleneck common in earlier YOLO variants, enabling efficient multi-stream inference.

For intra-camera tracking, we use ByteTrack [1], which associates both high- and low-confidence detections with existing tracks using a two-round matching scheme. This reduces identity loss from partial occlusions compared to trackers that discard low-score detections.

For cross-camera Re-ID, we use the Torchreid framework [10] with an OSNet-AIN backbone, which generates 512-dimensional appearance embeddings by learning features at multiple scales simultaneously.

B. Previous Research

N. Wojke et al. [2] proposed DeepSORT, which extends SORT by adding a deep appearance descriptor and Mahalanobis distance metric for data association, establishing the standard tracking-by-detection paradigm.

Y. Zhang et al. [3] presented StrongSORT, which improves upon DeepSORT by using a stronger ReID backbone and an EMA-based embedding update strategy, achieving consistent gains on MOT17 and MOT20 benchmarks.

V. Gautam et al. [4] proposed YOLORe-IDNet, which combines detection and ReID into a single forward pass. While efficient, its tightly coupled design limits modularity and does not support geometry-based identity recovery in overlapping zones.

C.-C. Wang et al. [5] addressed track fragmentation in crowded scenes by applying Gaussian Mixture Models to split tracklets containing multiple identities. This post-hoc correction is complementary to our pipeline, which prevents identity mixing earlier through spatial matching.

Z. Wu et al. [6] incorporated 3D geometric constraints into multi-camera tracking as a fallback when appearance features are unreliable. Our spatial matching module adopts a similar motivation, but relies on 2D homography matrices that are simpler to compute.

R. Hachiuma et al. [7] showed that spatial trajectories are more reliable than visual features in appearance-constrained settings such as operating rooms. This finding motivated our design choice to prioritize geometric matching in overlapping camera zones.

III. Methodology

This section describes the system pipeline and each of its modules.

A. System Overview

As shown in Fig. 1, the pipeline consists of four stages: (1) Person Detection, (2) Intra-Camera Tracking, (3) Spatial Matching via Homography, and (4) Cross-Camera Re-Identification. The backend is built with FastAPI and runs as an asynchronous service. Each camera feed is processed by an independent worker, and a central coordinator aggregates the results and maintains the global identity registry. Models are shared across workers to minimize GPU memory usage.

Tracking results are delivered to operator clients through two synchronized channels. Raw video frames are streamed via MJPEG, while structured tracking data—bounding boxes, identity labels, and floor plan positions—is pushed through WebSocket connections. The frontend overlays WebSocket data onto the MJPEG stream in real time.

Fig 1: System Architecture

Fig. 1. Overview of the proposed multi-camera tracking pipeline.

B. Person Detection

We use YOLOv26m [9] as the person detector. Input frames are resized to 480×480 pixels and normalized. The model uses an NMS-free decoupled head design, which reduces per-frame inference time. We apply a confidence threshold of 0.3 to capture detections that would otherwise be filtered by standard NMS-based thresholds.

C. Intra-Camera Tracking

Bounding boxes are passed to ByteTrack [1] for multi-object tracking within each camera. ByteTrack assigns a local Track ID to each person and maintains it across frames using Kalman filtering and IoU-based matching, with a 20-frame buffer for temporarily occluded tracks.

D. Spatial Matching and Mapping

The spatial matching module resolves identities geometrically for cameras with overlapping fields of view. We compute a 3×3 homography matrix $\mathbf{H}$ for each camera using RANSAC over manually defined calibration point pairs. During tracking, the bottom-center point of each bounding box, approximating the person's foot position (x_c, y_{max}), is projected onto the shared floor plan:

$$\lambda \mathbf{p}_{map} = \mathbf{H}\mathbf{p}_{img} \tag{1}$$

where λ is a scaling factor. The system then constructs a pairwise Euclidean distance matrix over all projected positions across camera views and applies the Hungarian algorithm to find optimal identity assignments. Pairs within 100 pixels (campus) or 200 pixels (factory) are merged under a single Global ID.

Fig 2: Homography Mapping

Fig. 2. Homography transformation mapping a camera view to a 2D floor plan using calibrated correspondence points.

E. Cross-Camera Re-Identification

For non-overlapping camera pairs, spatial proximity cannot resolve identities. We define handoff zones as a 5% margin along each edge of every camera frame. When a tracked person enters these zones, the system triggers the Re-ID module to extract a 512-dimensional appearance embedding using the Torchreid framework [10] with an OSNet-AIN backbone (yielding an average 68.7% Rank-1 and 62.0% mAP on MTMMC). Embeddings are L_2-normalized and stored in a FAISS index [11] with a 60-second expiry. Incoming edge detections are matched against this registry using cosine similarity; tracks exceeding a threshold of 0.70 inherit the matched Global ID.

Fig 3: Handoff Zones

Fig. 3. A 5% edge margin on each camera frame triggers Re-ID for cross-camera association in non-overlapping zones.

IV. Experimental Results

This section presents the results of our evaluation. We tested the system on the MTMMC dataset [8].

A. Dataset and Experimental Setup

We evaluated our system on the MTMMC (Multi-Target Multi-Modal Camera) dataset [8], which provides synchronized video feeds from two environments: a university campus with outdoor pathways and an indoor factory with workstations. The campus environment utilizes four tracking cameras that exhibit viewpoint variation, natural lighting changes, and occlusions. The factory environment also utilizes four cameras, presenting distinct challenges such as uniform worker attire and repetitive background patterns.

All experiments were conducted on an NVIDIA GeForce RTX 4060 GPU (8 GB VRAM). The YOLOv26m detector was accelerated using NVIDIA TensorRT with an input resolution of 480×480 pixels and a confidence threshold of 0.3. ByteTrack was configured with a 20-frame track buffer. For cross-camera matching, the Re-ID similarity threshold was set to 0.70, and the FAISS index utilized Inner Product search for 512-dimensional embeddings. To maintain real-time performance across all concurrent streams, the system was configured to process every third frame, yielding an effective processing rate of 7.67 FPS per camera.

B. Overall Tracking Performance

Table I presents the quantitative tracking evaluation across both environments. We evaluated the Multiple Object Tracking Accuracy (MOTA), IDF1, ID Switches, Precision, and Recall for each camera feed.

TABLE I
CONSOLIDATED TRACKING PERFORMANCE (TENSORRT OPTIMIZED, FRAME-SKIP = 2)

Camera	mAP@50	Recall	Precision	MOTA	MOTP	IDF1	ID Sw.	Re-ID mAP	Rank-1
Factory Environment									
c09	74.5%	82.2%	82.7%	63.9%	9.4%	68.6%	6	71.6%	78.3%
c12	63.4%	74.8%	75.6%	52.4%	15.5%	62.8%	1	60.1%	66.8%
c13	67.2%	78.6%	78.3%	57.4%	13.8%	62.8%	1	65.1%	71.8%
c16	68.6%	77.3%	82.3%	59.7%	11.5%	58.9%	42	67.4%	74.1%
Factory Avg.	**68.4%**	**78.2%**	**79.7%**	**58.4%**	**12.5%**	**63.3%**	**50**	**66.1%**	**72.8%**
Campus Environment									
c01	70.3%	82.1%	75.7%	57.4%	19.3%	59.0%	48	65.1%	71.8%
c02	56.4%	71.7%	72.2%	47.2%	18.7%	60.7%	3	54.9%	61.6%
c03	50.7%	69.1%	66.5%	39.8%	23.4%	55.3%	15	47.5%	54.2%
c05	65.0%	77.1%	78.0%	56.1%	17.0%	65.7%	0	63.8%	70.5%
Campus Avg.	**60.6%**	**75.0%**	**73.1%**	**50.1%**	**19.6%**	**60.2%**	**66**	**57.8%**	**64.5%**

C. Processing Latency

We measured the processing latency per frame for each pipeline stage using TensorRT GPU acceleration. Table II shows the average latency breakdown.

TABLE II
AVERAGE PER-FRAME PROCESSING LATENCY (TENSORRT GPU)

Pipeline Stage	Avg. Latency (ms)
Object Detection (YOLOv26m TensorRT)	11.9
Data Association (ByteTrack)	1.2
Feature Extraction (OSNet)	36.3
Total End-to-End Processing	**49.4**
Estimated FPS Cap	**20.3 FPS**

D. Qualitative Results

Fig. 4 shows sample tracking outputs from the system. The results demonstrate that the system can maintain consistent identity assignment across camera views.

Fig. 5 compares the IDF1 and MOTA scores across different pipeline configurations to show the contribution of each module.

Fig. 6 shows the operator dashboard, where tracked individuals from all cameras are displayed on the shared 2D floor plan.

E. Discussion

The factory environment achieved higher average MOTA (58.4%) and IDF1 (63.3%) than the campus environment (50.1% and 60.2%, respectively). The campus presented greater challenges due to viewpoint variation, outdoor lighting changes, and occlusions from vegetation, which degraded detection recall and identity preservation.

To assess module contributions, we evaluated four configurations: (a) detection and tracking only, (b) with Re-ID added, (c) with spatial matching added, and (d) the full pipeline. As shown in Fig. 5, spatial matching provides a larger IDF1 improvement than Re-ID alone in the campus environment, where camera views overlap. Re-ID contributes more in the factory, where views are disjoint. The full pipeline achieves the best results in both environments.

The system has two main limitations. First, tracking accuracy is sensitive to the quality of manual homography calibration. Second, detection reliability degrades in highly crowded scenes with persistent occlusions.

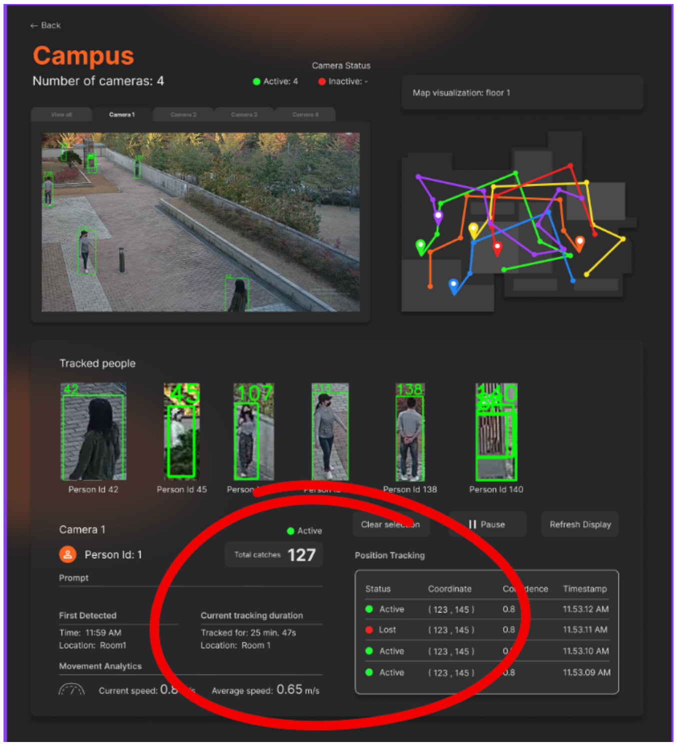

Fig. 4. Sample multi-camera tracking output showing consistent global identity assignment across camera views.

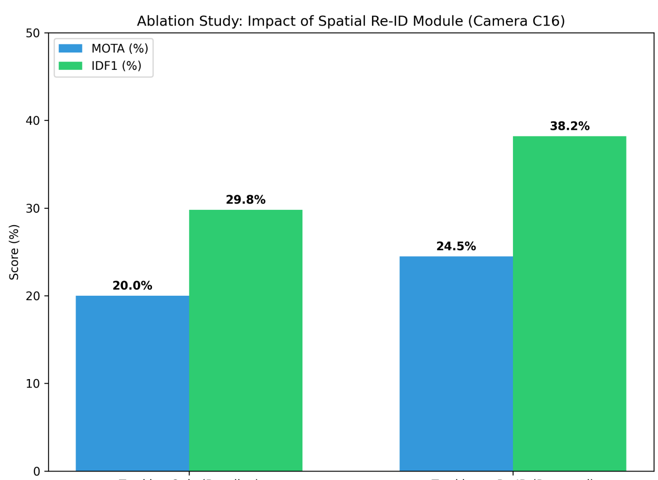

Fig. 5. Comparative IDF1 and MOTA scores across ablation configurations: (a) Detection + Tracking only, (b) with Re-ID only, (c) with Spatial Matching only, and (d) full pipeline.

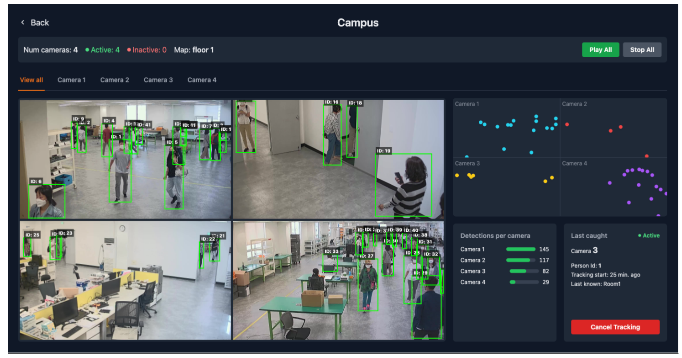

Fig. 6. The SpotOn operator dashboard displaying real-time person locations on a unified 2D floor plan, with color-coded identity markers aggregated from multiple camera feeds.

V. Conclusion

In this paper, we presented SpotOn, a multi-camera person tracking system that uses homography-based spatial matching as the primary identity resolution method for overlapping camera views, reserving appearance-based Re-ID for non-overlapping blind spots. Evaluated on the MTMMC dataset, the system achieved an average IDF1 of 61.8% and MOTA of 54.3% with a per-frame latency of 49.4 ms across four concurrent camera streams per environment.

For future work, we plan to automate the homography calibration process, which currently requires manual point selection. We also intend to explore temporal motion prediction for handling long occlusions and to validate the system on additional benchmarks and real-world deployments.

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